

OBRESTAD, Norway

WMO: 014120

Lat: 58.6585N

Lon: 5.5546E

Elev: 85

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.4	22.1	6.7	7.8	22.5	11.5	9.9	25.2	45.4	38.6	41.3	39.9	8.3	70	0.832

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.3	70.6	61.7	67.6	60.3	65.4	59.6	63.6	68.3	62.1	66.2	60.7	64.4	12.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.5	81.9	66.1	60.2	78.0	64.5	58.8	74.2	63.0	29.0	68.6	27.9	66.2	26.9	64.6	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 40.9	35.9	32.1	17.3	78.7	4.7	3.0	13.8	80.9	11.1	82.7	8.4	84.4	5.0	86.6		
(5)			16.0	66.9	5.0	2.1	12.4	68.5	9.4	69.7	6.6	70.9	2.9	72.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.5	36.2	35.2	37.3	43.2	48.4	53.1	58.4	59.3	55.0	48.9	42.9	39.3		
	DBStd	9.77	6.51	5.76	5.18	4.48	5.24	4.25	4.55	3.73	4.00	4.74	6.31	6.67		
	HDD50	2184	427	413	393	211	92	13	0	0	7	77	220	331		
	HDD65	6769	892	833	858	654	516	357	213	184	301	501	664	796		
	CDD50	909	0	0	0	7	42	107	262	287	157	42	5	0		
	CDD65	18	0	0	0	0	1	1	9	6	1	0	0	0		
	CDH74	31	0	0	0	0	3	1	19	8	0	0	0	0		
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0		
	WSAvg	15.5	18.4	16.1	14.6	13.9	14.6	13.9	13.3	14.3	15.9	16.8	16.9	17.9		
	PrecAvg	60.5	6.2	4.8	4.1	3.3	2.8	3.1	4.1	5.8	6.2	7.0	6.7	6.3		
(15) Precipitation	PrecMax	75.6	11.2	10.4	9.5	6.0	5.6	6.1	8.5	11.3	10.3	13.1	12.2	11.9		
	PrecMin	40.9	0.2	1.2	0.6	1.4	0.4	0.8	1.3	0.8	1.6	2.3	1.9	0.2		
	PrecStd	8.6	2.6	2.5	2.0	1.1	1.3	1.5	1.8	2.1	2.2	3.0	3.0	3.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.5	46.4	49.8	60.7	69.0	70.1	76.3	74.5	68.4	60.3	54.8	50.9		
		MCWB	46.1	44.6	43.0	50.5	57.2	59.8	64.3	59.7	59.2	55.9	52.6	48.8		
	2%	DB	46.7	44.6	45.7	55.9	64.4	65.8	71.4	69.7	64.6	58.4	53.5	49.4		
		MCWB	44.6	42.7	41.8	47.3	56.1	58.6	63.5	61.0	58.8	55.0	51.4	47.4		
	5%	DB	45.6	43.6	44.0	51.9	59.8	62.2	68.0	67.0	62.4	56.6	52.1	48.5		
		MCWB	43.5	42.0	41.5	46.2	52.7	57.1	62.1	60.0	57.9	53.8	49.8	46.4		
	10%	DB	44.6	42.6	43.0	49.3	55.7	59.3	65.2	65.1	60.6	55.3	50.8	47.2		
		MCWB	42.6	41.3	41.0	44.5	50.8	55.4	60.8	59.5	57.0	52.6	48.3	44.9		
	0.4%	WB	46.7	44.9	44.8	51.6	61.4	61.9	66.6	65.0	61.2	57.3	53.5	49.4		
		MCDB	47.9	45.9	47.7	58.6	68.3	66.8	73.7	69.4	65.1	59.3	54.5	50.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.3	43.6	43.1	48.6	56.5	59.8	64.5	63.0	59.9	56.0	52.0	48.2		
		MCDB	46.5	44.6	44.6	53.0	62.4	65.1	69.8	66.8	63.3	57.8	53.4	49.3		
	5%	WB	44.3	42.7	42.3	47.1	53.5	57.5	62.9	61.7	58.9	54.6	50.3	47.1		
		MCDB	45.6	43.6	43.6	50.7	57.8	61.6	67.5	65.3	61.7	56.4	51.8	48.3		
	10%	WB	43.1	41.7	41.5	45.6	51.0	55.6	61.2	60.5	57.6	53.3	49.0	45.6		
		MCDB	44.6	42.8	42.8	48.3	54.8	58.7	65.0	63.9	60.6	55.2	50.8	47.1		
	5% DB	MDBR	6.5	6.2	6.9	7.9	7.5	7.1	7.0	7.3	7.2	7.3	6.9	6.7		
		MCDBR	6.6	5.4	6.9	12.3	14.6	12.2	12.2	10.3	8.8	6.7	5.5	5.8		
		MCWBR	7.0	5.2	5.7	7.5	8.6	7.9	7.5	6.9	6.2	6.0	6.0	6.4		
		MCWBR	6.5	5.1	6.4	10.1	13.4	10.7	11.3	8.4	7.7	6.1	6.0	5.6		
(35) Mean Daily Temperature Range	5% WB	MCWBR	6.9	5.2	5.5	7.2	8.4	7.7	7.4	6.4	6.6	6.0	6.7	6.3		
	Clear-Sky Solar Irradiance	taub	0.314	0.336	0.354	0.385	0.372	0.366	0.379	0.381	0.363	0.343	0.317	0.292		
		taud	2.173	2.272	2.319	2.290	2.403	2.451	2.432	2.424	2.458	2.395	2.246	2.129		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	154	215	248	259	273	276	269	259	245	212	154	120		
		Edh at Noon	16	23	29	35	34	32	32	30	25	20	15	12		
(44) All-Sky Solar Radiation	RadStd		95	290	679	1223	1660	1781	1592	1230	792	378	133	62		
			14	46	68	112	172	193	153	100	68	41	18	7		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-209	N/A	N/A

Nomenclature: See separate page